AUTOMATED GENERATION OF ABSA MODELS FOR MEDICAL DEVICES USING SYNTHETIC DATA

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What is Aspect Based Sentiment Analysis?

- Aspect Based Sentiment Analysis identifies the sentiment expressed toward specific aspects of a product, instead of predicting only the overall review sentiment.

What we did:

- We used an ABSA-inspired multi-output approach: the model receives one review and predicts a score for every extracted aspect.

Review → Aspect Score Vector

Score meaning:

- positive mention = 1
- negative mention = -1
- aspect not mentioned = 0

Motivation & Project Definition

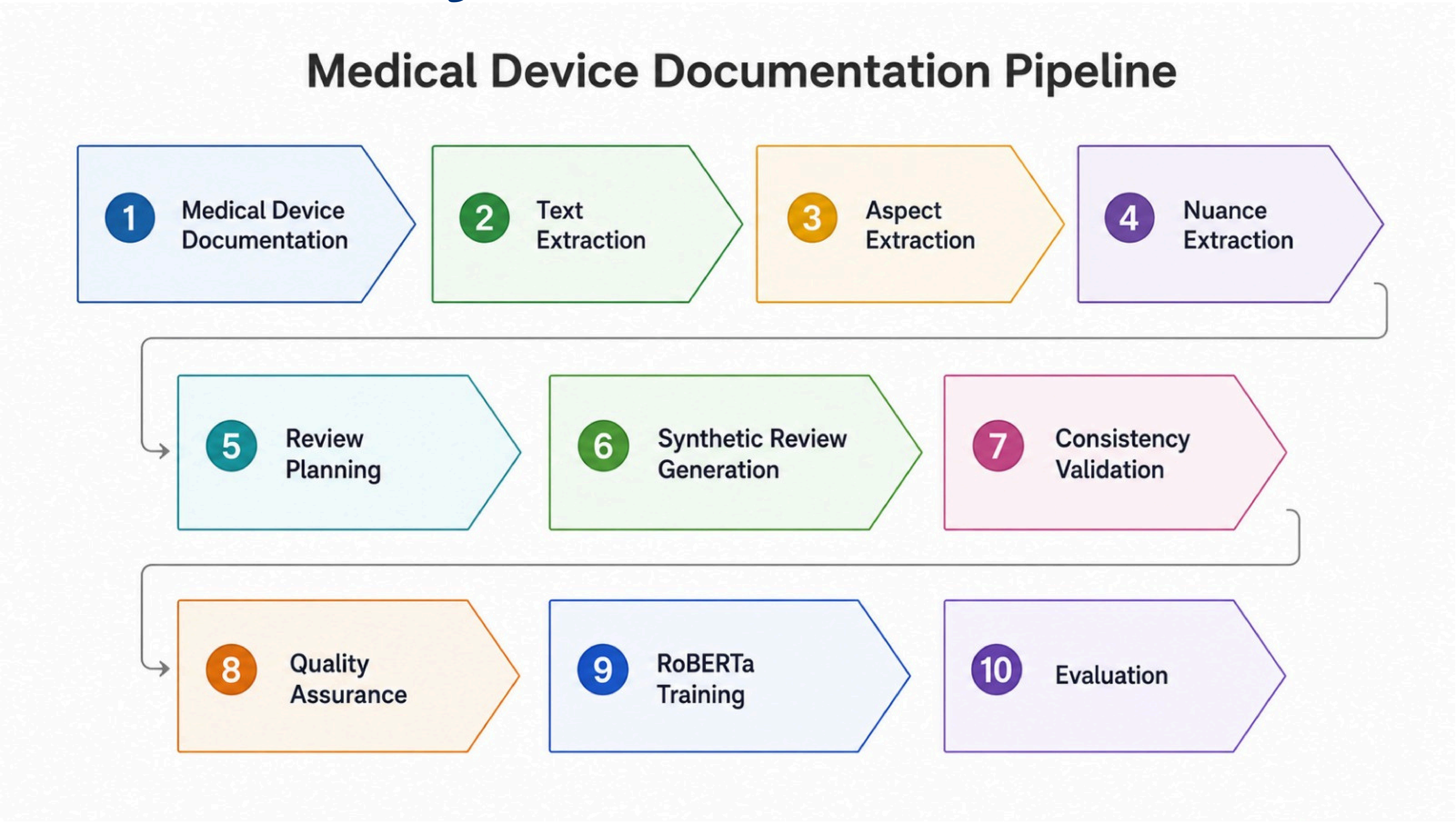
motivation

Medical device reviews contain valuable information, but building labeled datasets for sentiment analysis requires a lot of manual work. In addition, each medical device has different characteristics, making it difficult to reuse existing datasets and models.

Goal

The goal of this project is to develop a generic pipeline that automatically extracts product specific aspects from a medical device PDF, generates labeled synthetic reviews, validates data quality, and trains a sentiment prediction model.

Project Overview



PDF Processing

- **Input**

Medical Device Documentation

- **Processing Steps**

Text Extraction, Text Cleaning, Text Chunking (splitting large documents into smaller sections for LLM processing)

- **Output**

Structured text chunks ready for LLM processing

Automatic Aspect Extraction

- An aspect is a product feature that users can express an opinion about.
- Extracted automatically from each medical device documentation using an LLM.
- The prompt requires for 5-8 document-grounded, product-specific, non-overlapping aspects.
- Each product receives its own aspect list, based on its documentation.
- The extracted aspects are saved as a JSON schema with an aspect_name and a short description.
- This JSON schema is later used for synthetic review generation and model training.

Example Of Aspects

Uroject12

- Ease of Use
- Durability & Build Quality
- Cleaning & Maintenance
- Compatibility with Syringes
- Storage & Handling
- Performance in Fluid Delivery

Medtronic 53401

- Battery Life
- Pacing Modes
- Device Portability
- Output Adjustability
- Sensitivity & Detection
- Safety Features
- Rate Range

Nuance Attribute Extraction

- A nuance is a background factor that shapes how a synthetic review is written.
- Motivation: make generated reviews more diverse, natural, and realistic.
- A prompt is sent to the LLM to extract 6 nuance dimensions from the medical device documentation, with 3-6 possible values per dimension.
- Extracted nuances are saved as JSON: dimension_name, description, possible_values.
- Used later during review generation to guide the review's context, tone, and level of detail.

Example Of Nuance Attribute

Nuance Attribute	Example Value	Example Review Phrase
Experience Level	Novice	"As a first-time user..."
Reviewer Profession	Nurse	"In our clinic..."
Usage Context	Emergency Room	"During emergency procedures..."
Review Type	User Experience	"After several weeks of use..."
Review Perspective	End User	"I found the device..."
Review Detail Level	Detailed	"The device performed consistently throughout testing..."

Review Generation Controls

Example Aspect: cleaning and maintenance

Sentiment	Example Review
Positive (+1)	"The cleaning process for this device is straightforward and effective, making maintenance simple even for beginners."
Negative (-1)	"The device performs fluid delivery smoothly and efficiently, making syringe use straightforward. However, cleaning is quite challenging due to its intricate parts, which makes maintenance frustrating."
Not Mentioned (0)	"The device is difficult to operate smoothly, and the lever mechanism requires excessive force, making it frustrating to use."

Model Architecture RoBERTa

Why RoBERTa-base?

- Pretrained transformer model for NLP tasks
- Strong performance in sentiment analysis
- Fine tuned on the generated medical review dataset

Training Process

- Start from the validated synthetic review dataset
- Split into Train / Validation / Test
- Fine-tune RoBERTa
- Input: review text
- Output: continuous aspect-score vector
- Task: multi-output regression
- Evaluate model performance

Fine-Tuning Strategy

- Loss Function: Mean Squared Error (MSE)
- Learning Rate: 1e-5
- Batch Size: 8
- Epochs: 3
- Weight Decay: 0.05
- Maximum Sequence Length: 256
- Hidden Dropout: 0.2
- Attention Dropout: 0.2
- Classifier Dropout: 0.2
- Dynamic Padding (DataCollatorWithPadding)
- Early Stopping (Patience = 1)
- Best Model Selection using Validation Loss

Validation

- JSON validation: used Python json library to parse each JSONL row and check required fields
- Duplicate detection: used Python Counter to detect repeated review texts and remove duplicates
- Label consistency checks: custom Python rules: validated -1 / 0 / 1 scores and selected/unselected aspects

Dataset	Reviews	Invalid Rows	Duplicates
Uroject12	1960	0	40 removed (final 0)
Medtronic 53401	1980	0	20 removed (final 0)

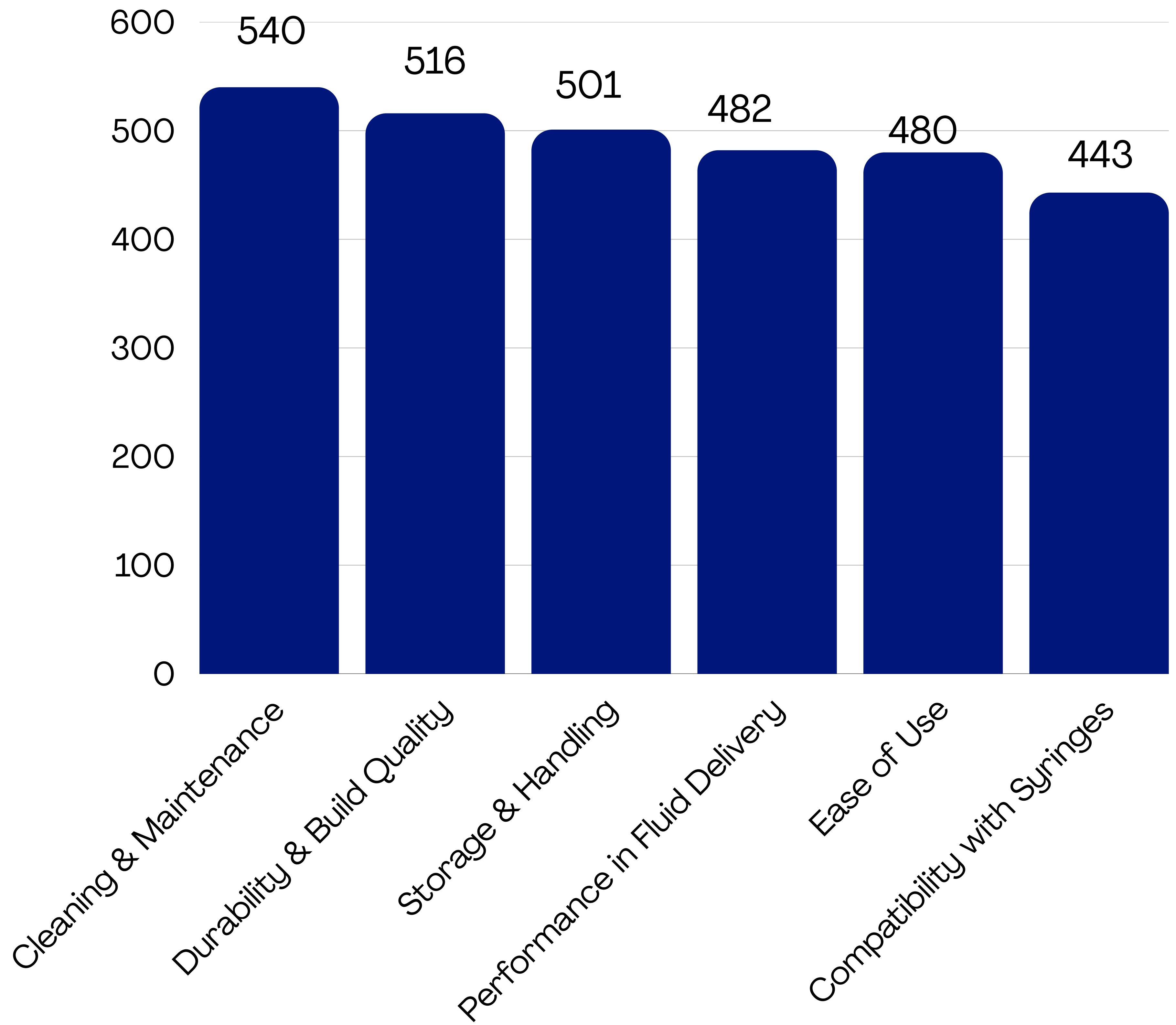
Validation

- Aspect Distribution Analysis: used Python Counter to count aspect frequency and sentiment distribution.
- Manual Review Sampling: used pandas to export a random sample of 100 reviews to CSV for human inspection.
- LLM Consistency Audit: used an LLM verifier through the OpenAI API to check whether review text matches the assigned aspect labels.

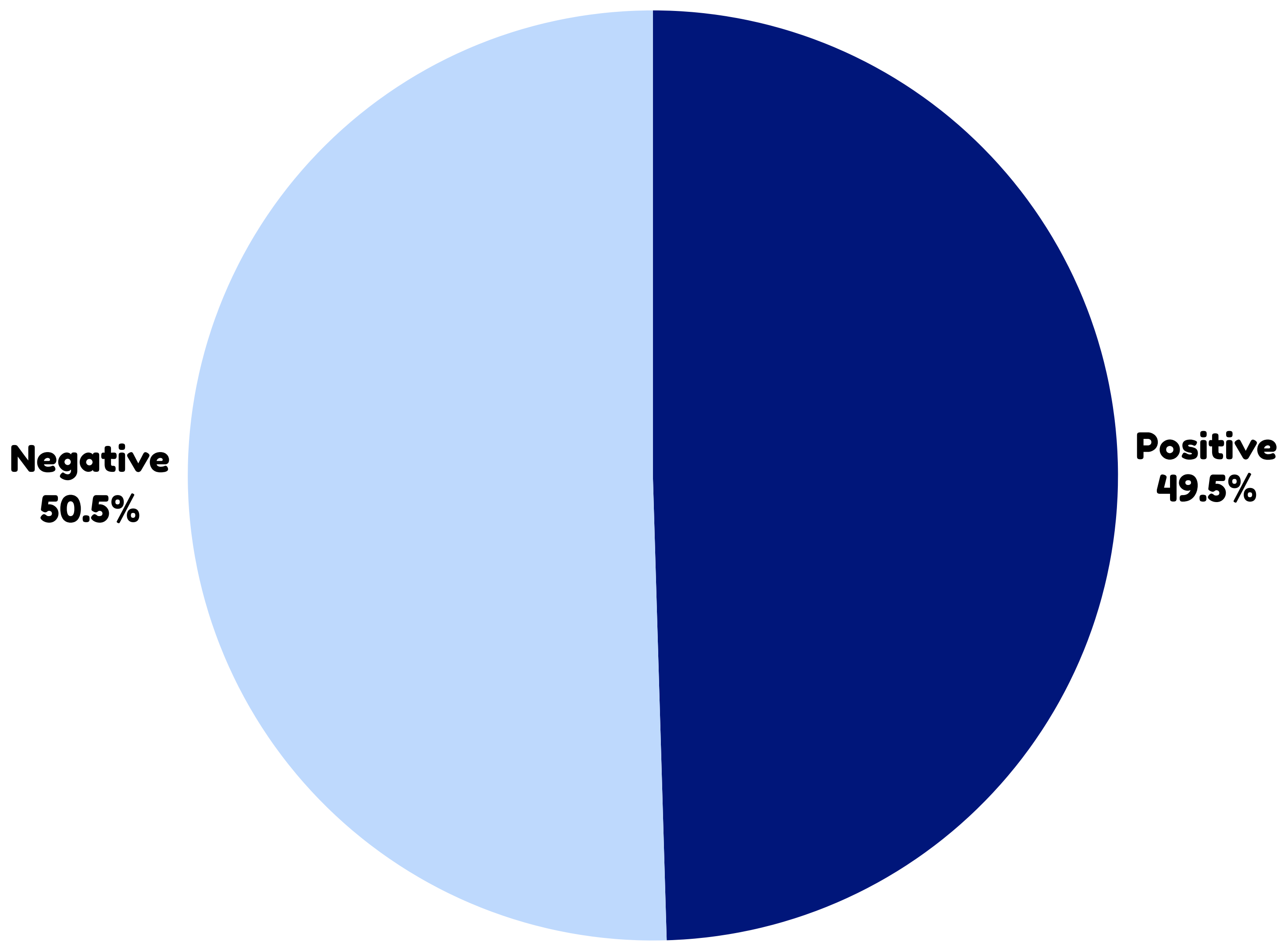
Dataset	Manual Review Sampling	Consistency Rate
Uroject12	98%	99.17%
Medtronic 53401	99%	96.79%

Uroject12

Aspect Distribution

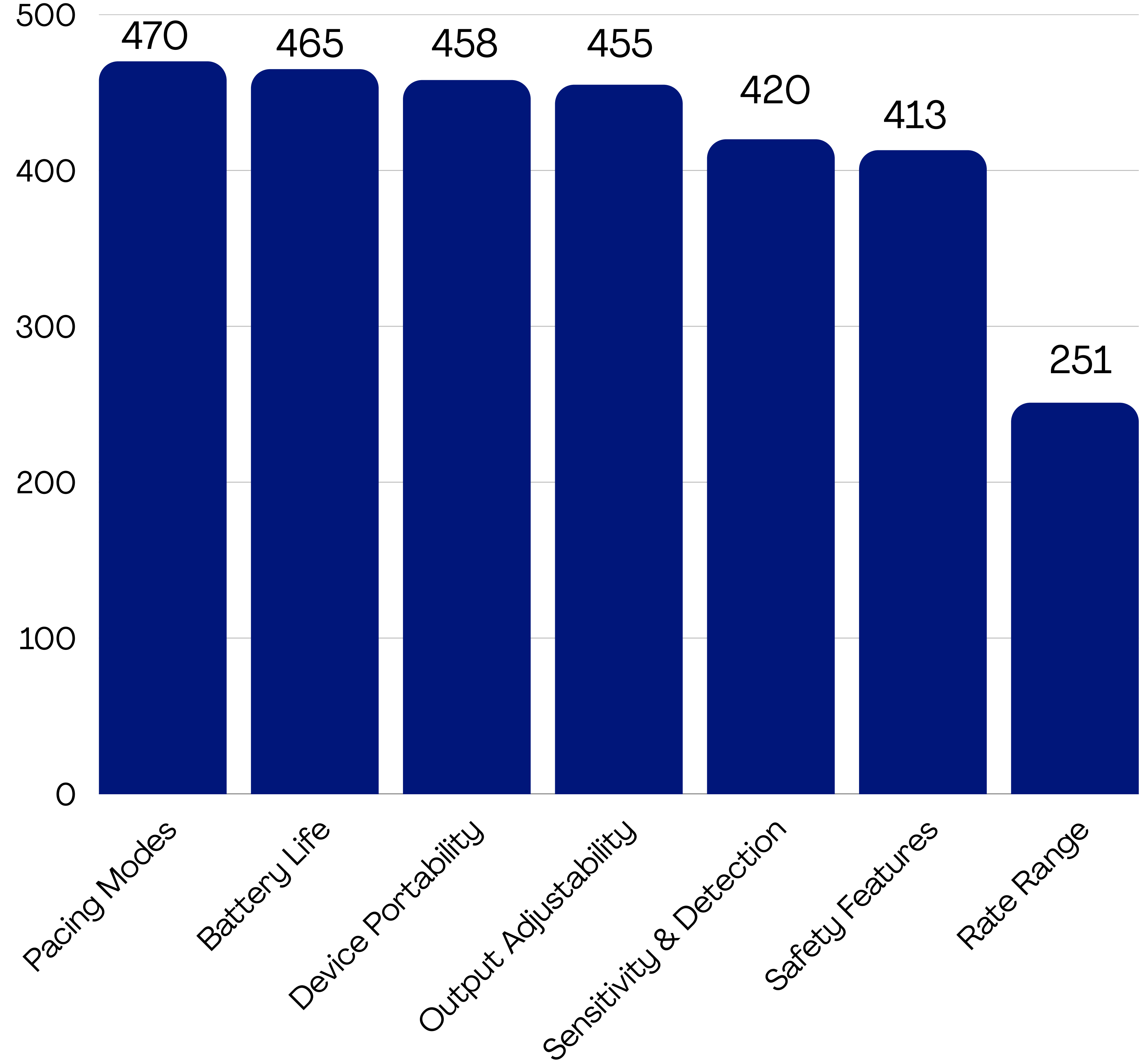


Overall Sentiment Distribution

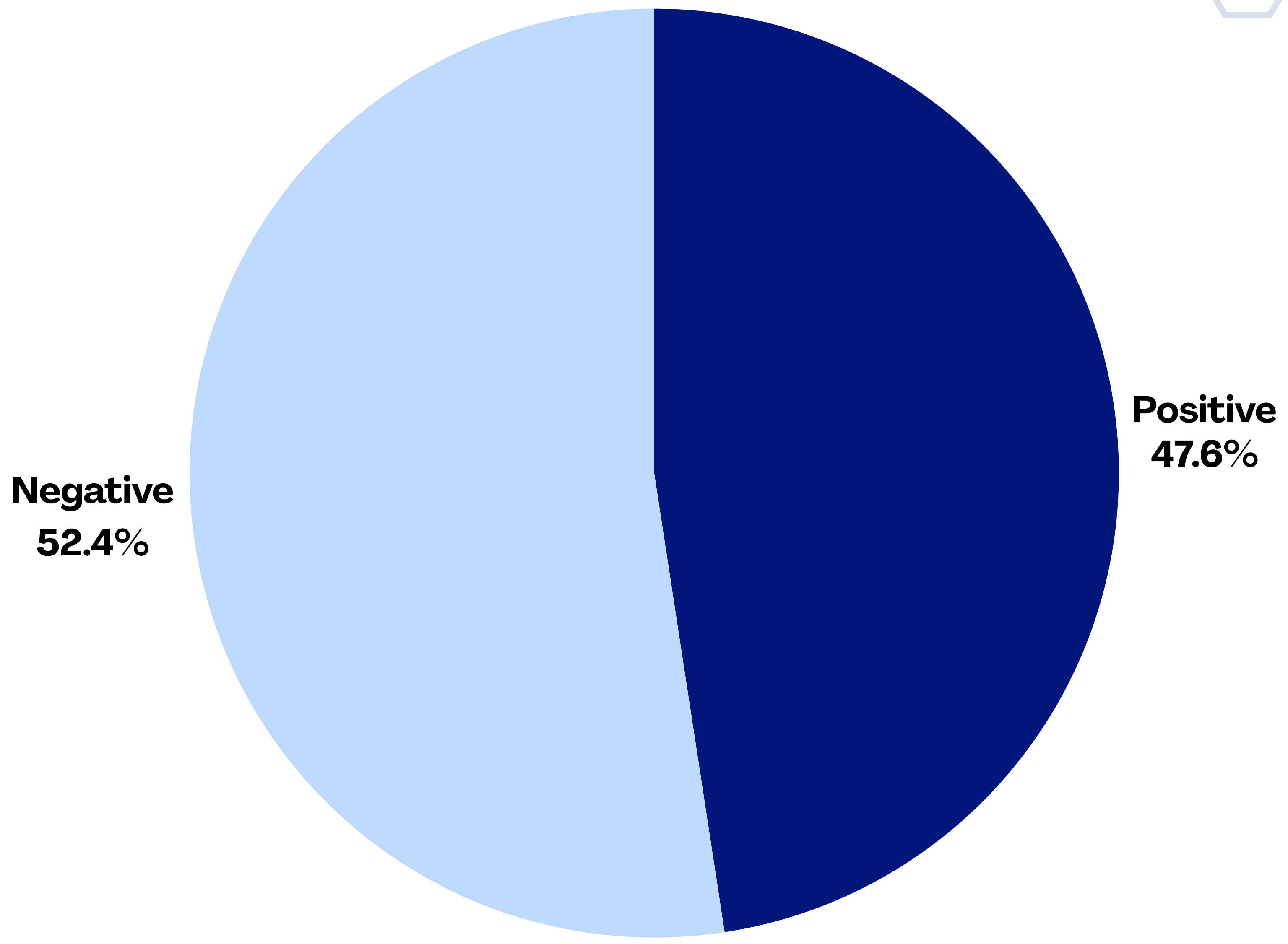


Medtronic 53401

Aspect Distribution



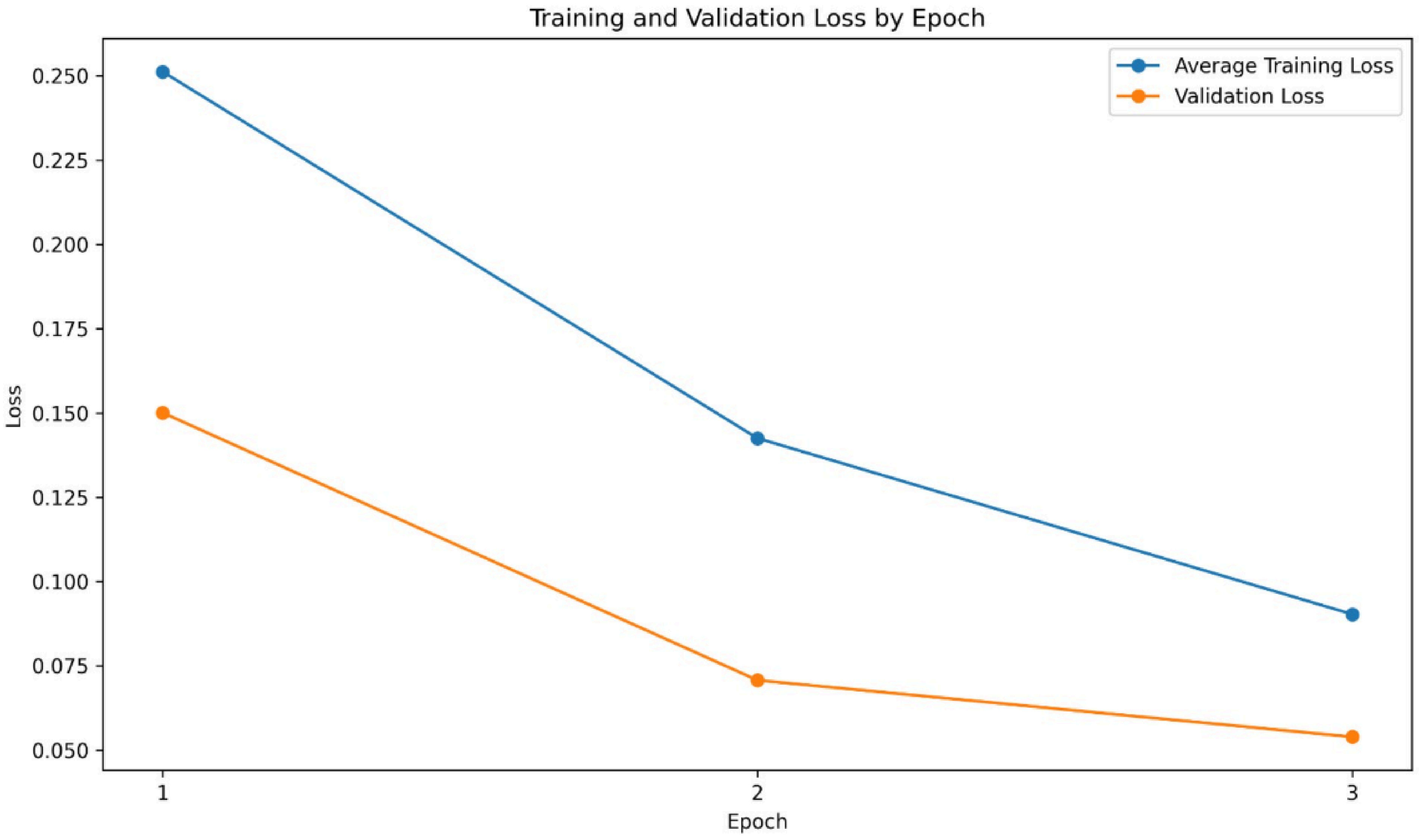
Overall Sentiment Distribution

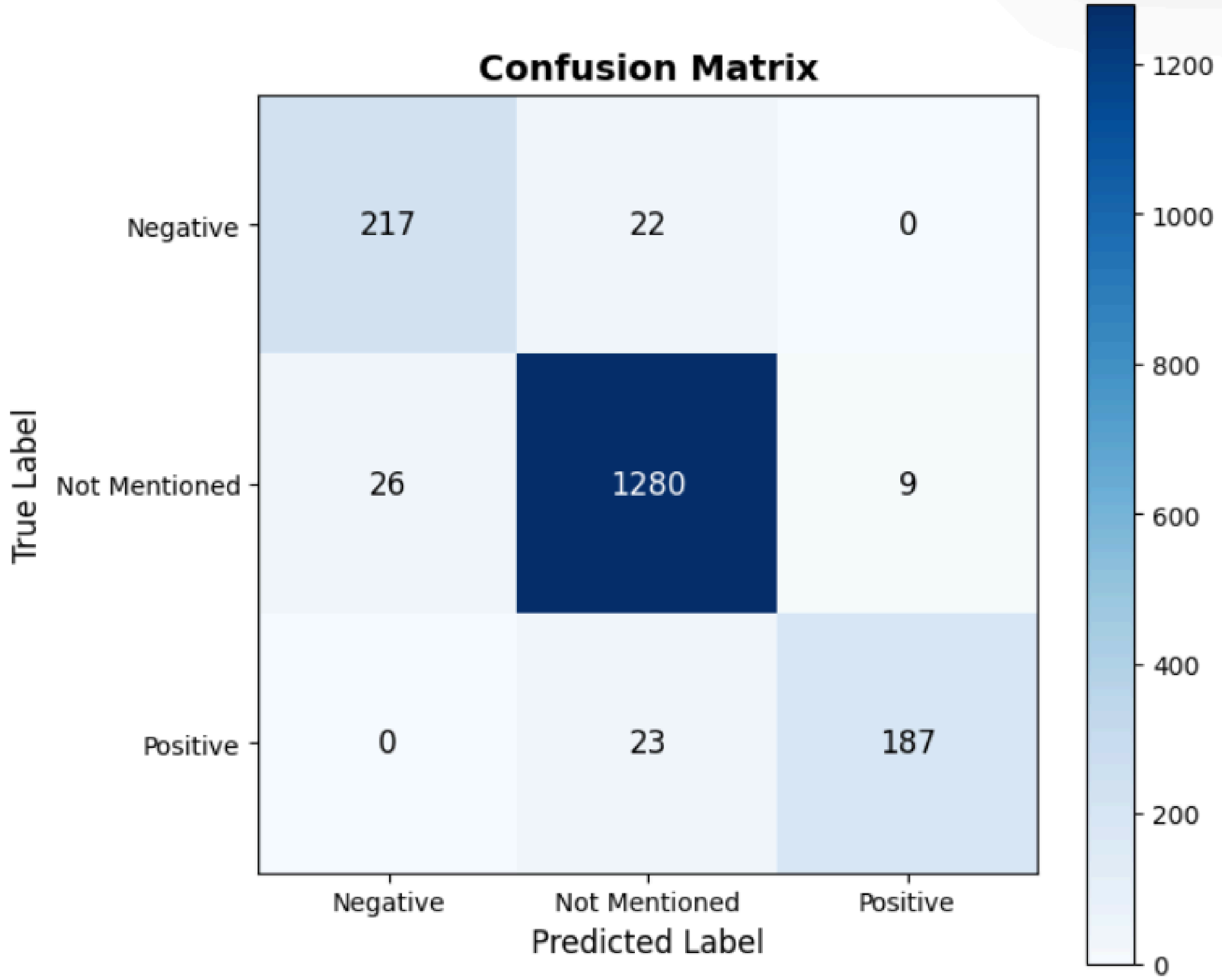


Results 1 : Uroject12

Overall Performance

95.5%	78.2%	93.0%	93.8%	92.4%
Element Accuracy	Exact Vector Match	Macro F1	Precision	Recall

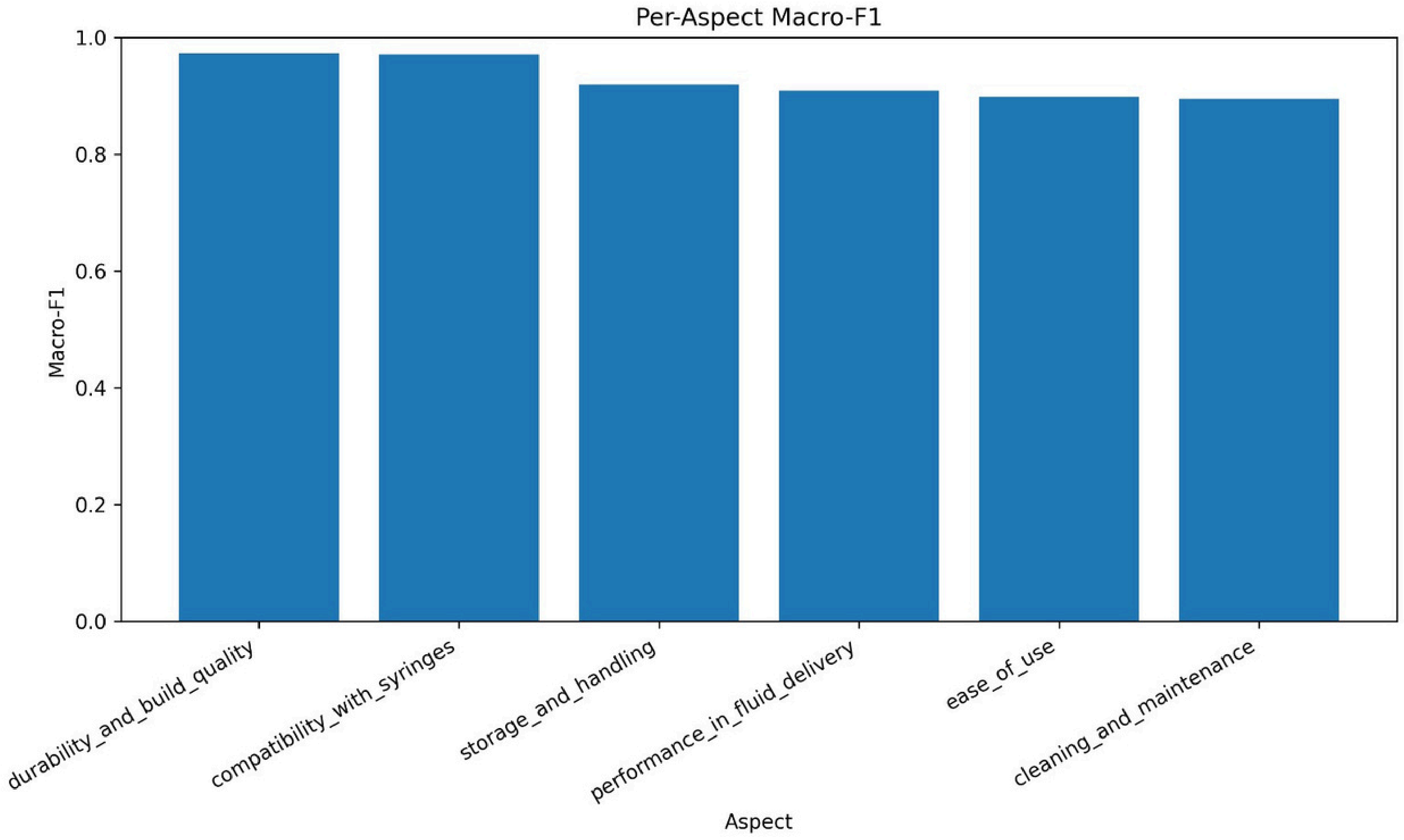




Results 1 : Uroject12

Per-Aspect Macro F1

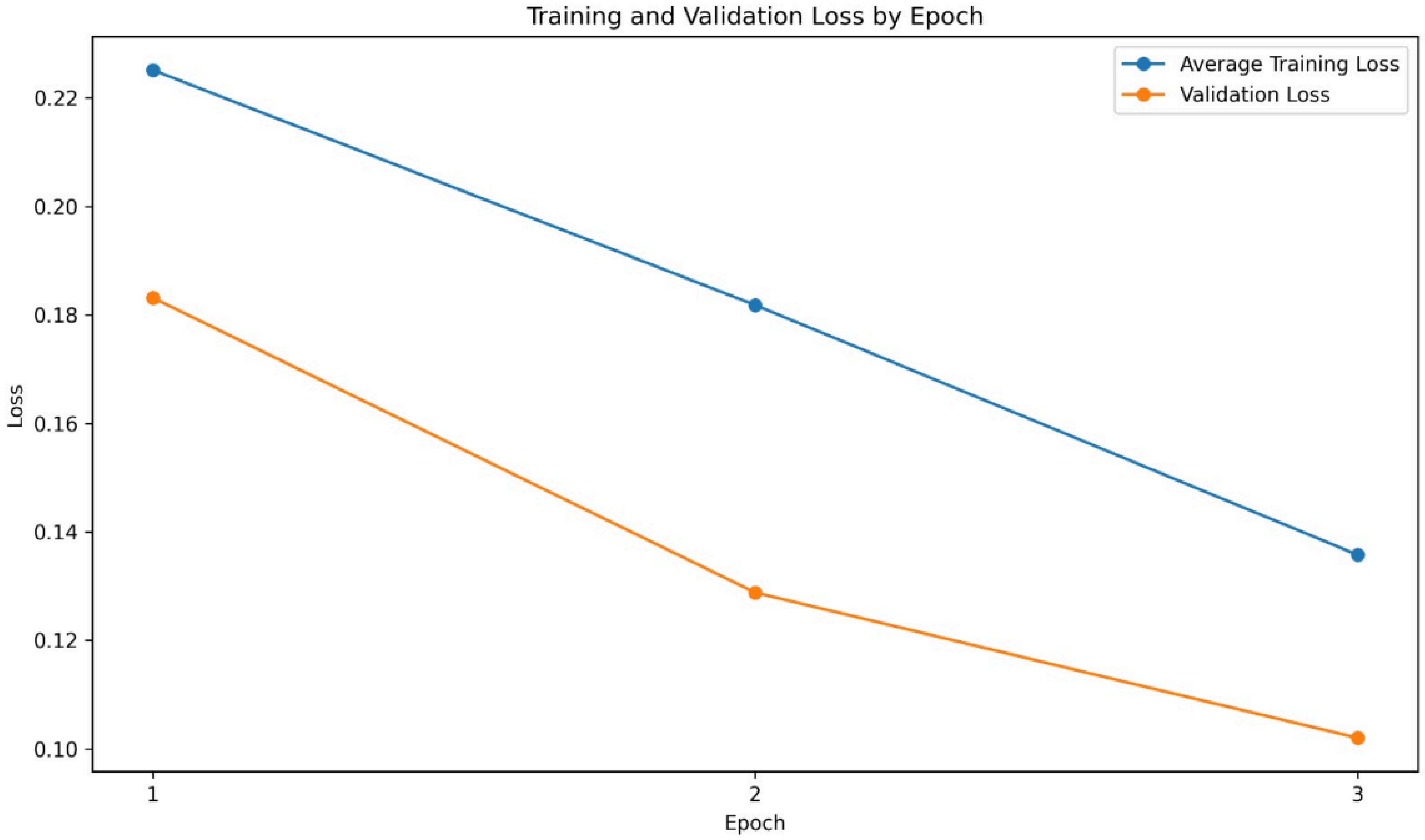
89.8%	97.3%	89.5%	97.1%	92.0%	90.8%
Ease of Use	Durability & Build Quality	Cleaning & Maintenance	Compatibility with Syringes	Storage & Handling	Performance in Fluid Delivery

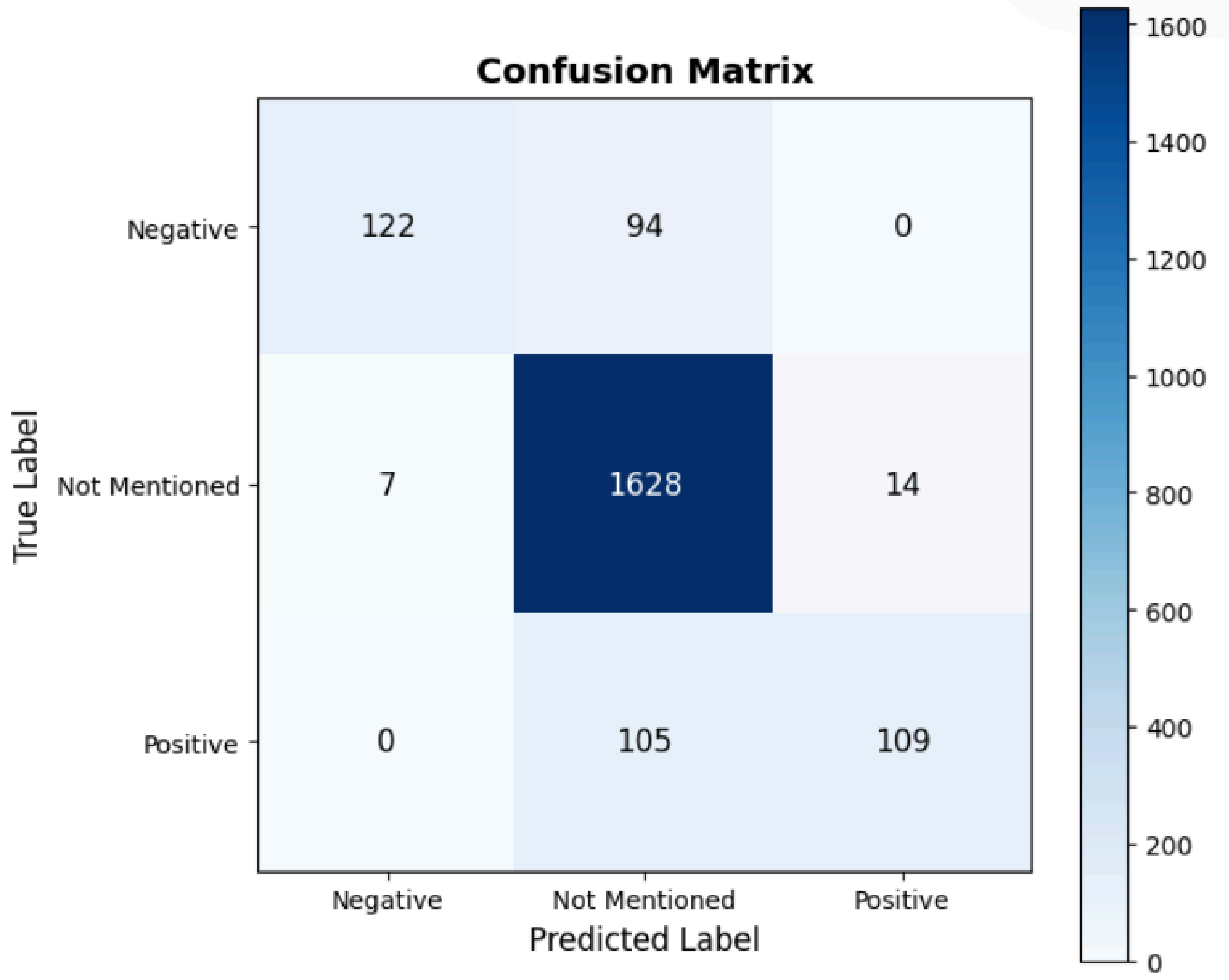


Results 1 : Medtronic 53401

Overall Performance

89.4%	40.4%	76.4%	90.8%	68.7%
Element Accuracy	Exact Vector Match	Macro F1	Precision	Recall

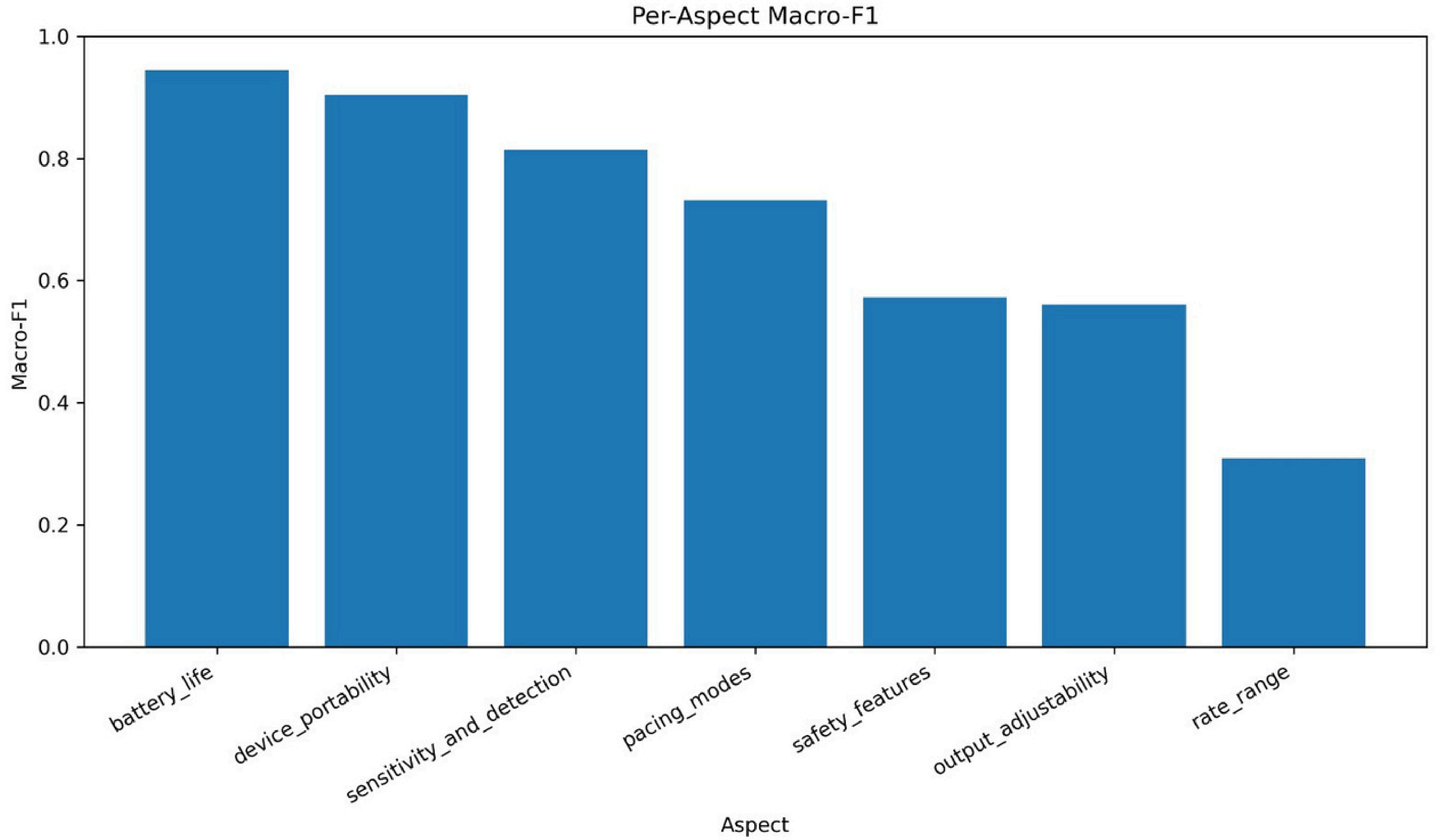




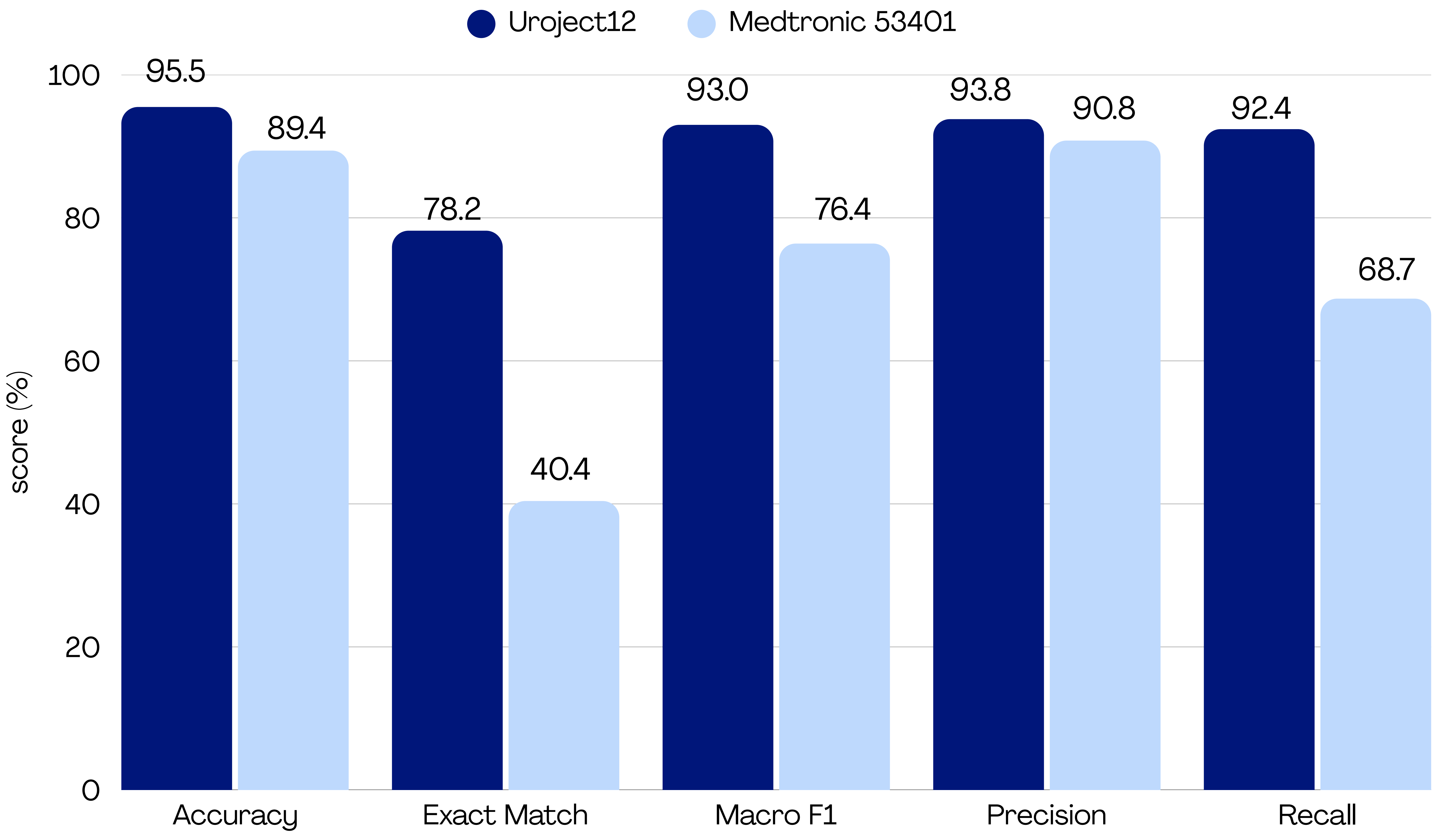
Results 1 : Medtronic 53401

Per-Aspect Macro F1

94.4%	90.5%	81.4%	73.1%	57.3%	56.1%	30.9%
Battery Life	Device Portability	Sensitivity & Detection	Pacing Modes	Safety Features	Output Adjustability	Rate Range



Comparison Between Devices



Model Architecture DeBERTa

Why DeBERTa-v3-small?

- Advanced transformer architecture with disentangled attention.
- Improved contextual understanding compared to RoBERTa.
- Efficient performance with fewer parameters.
- Well suited for aspect-based sentiment analysis in medical reviews.

Training Process

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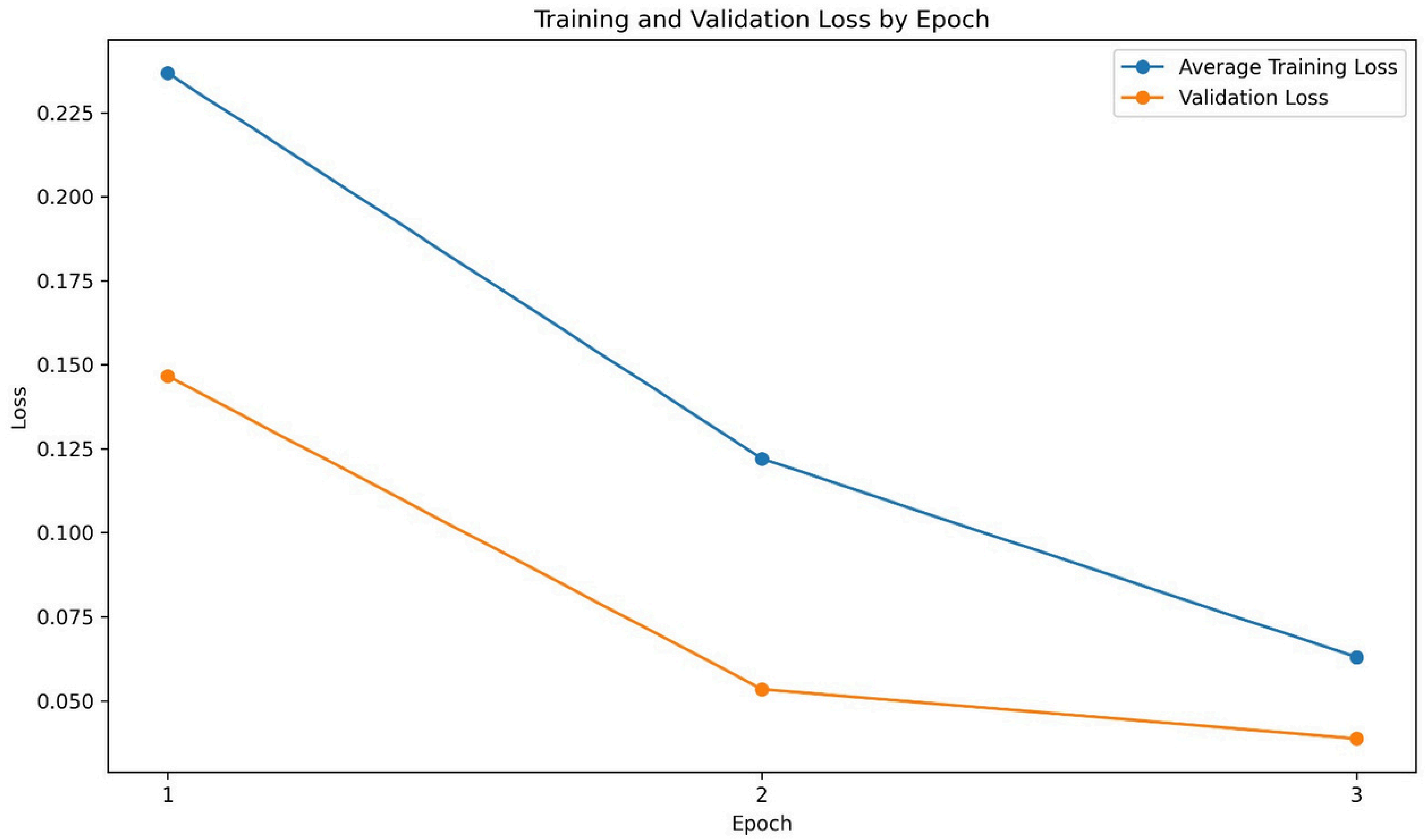
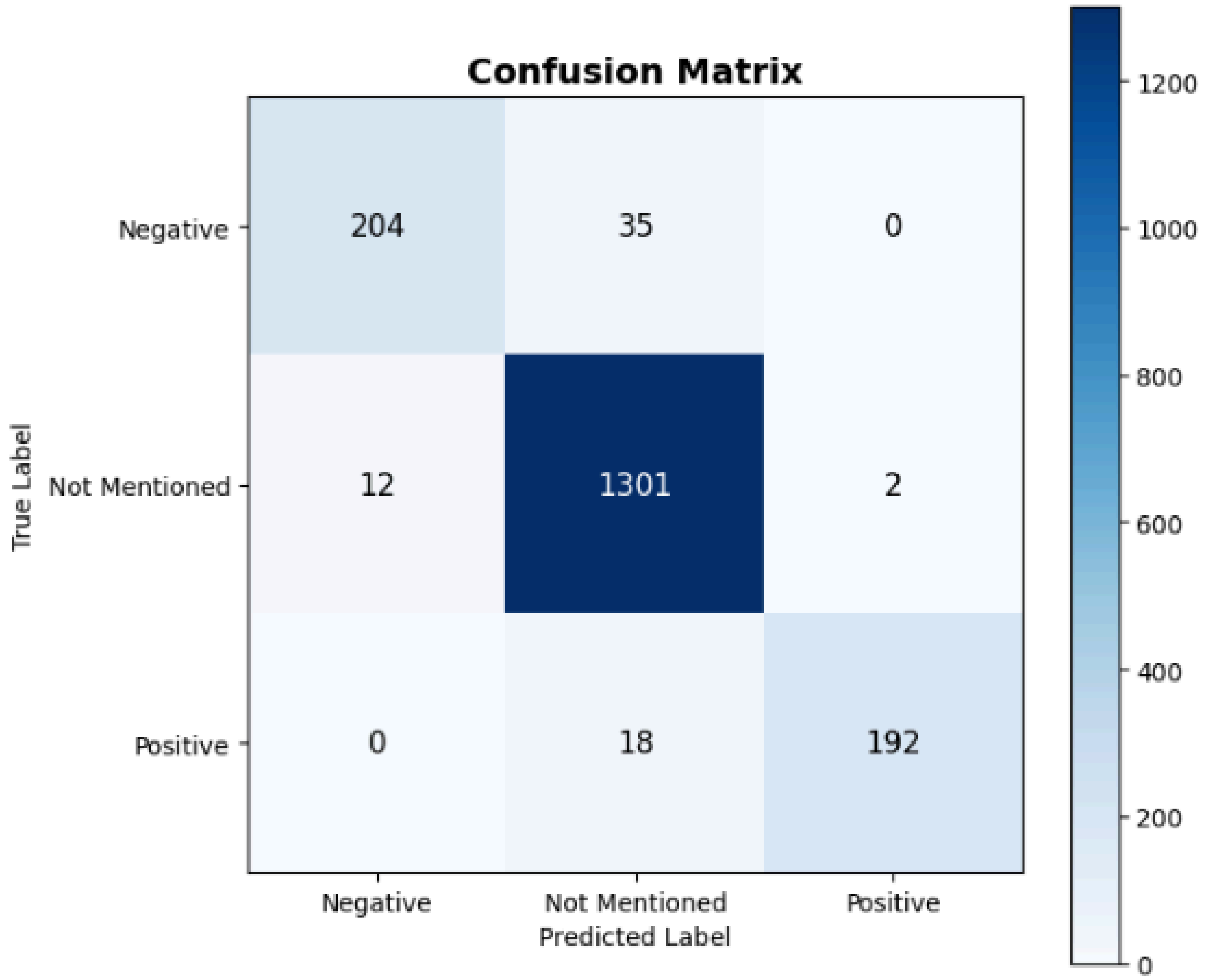
Results 2 : Uroject12

Overall Performance

96.2%	79.9%	94.1%	96.5%	91.9%
Element Accuracy	Exact Vector Match	Macro F1	Precision	Recall

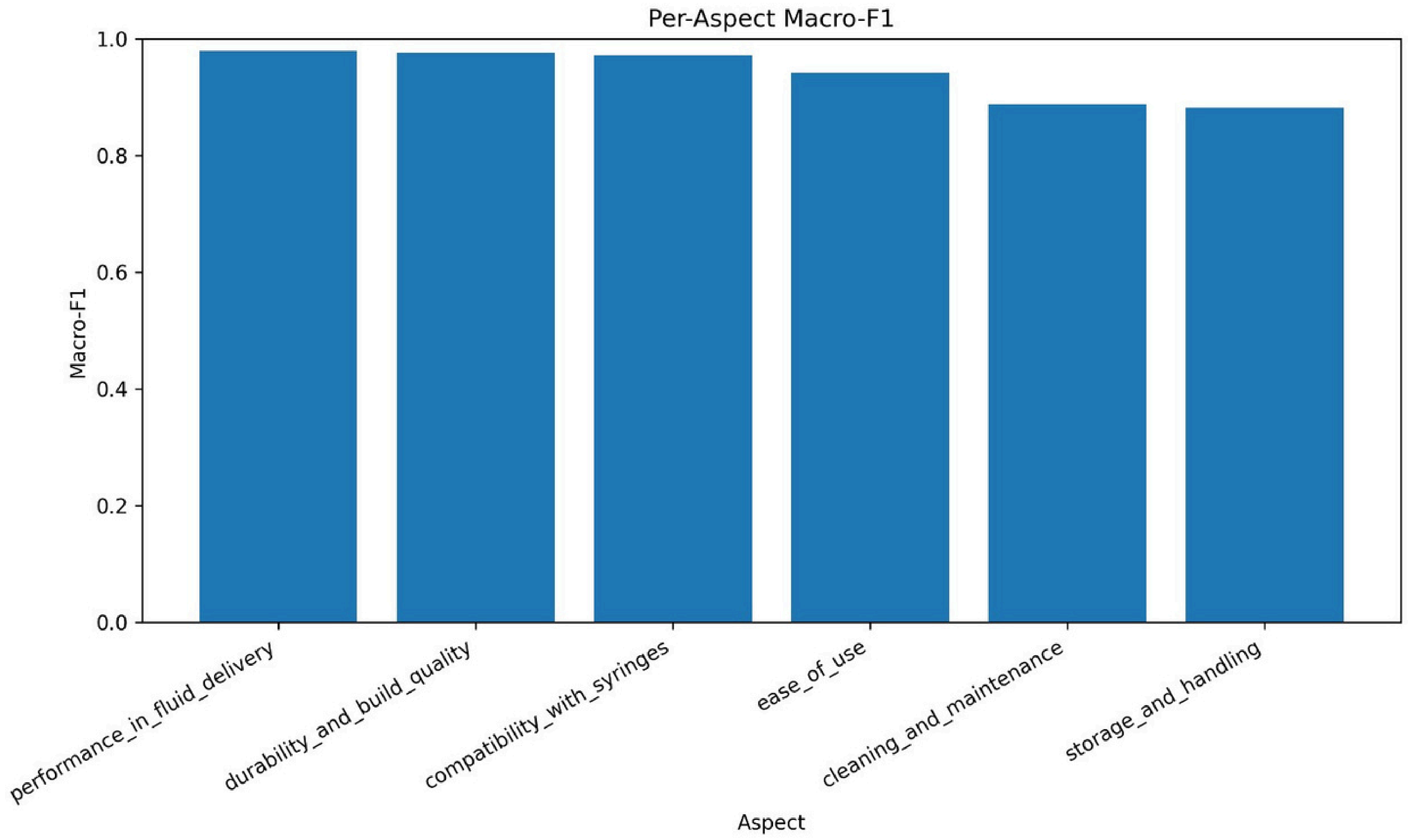




Results 2 : Uroject12

Per-Aspect Macro F1

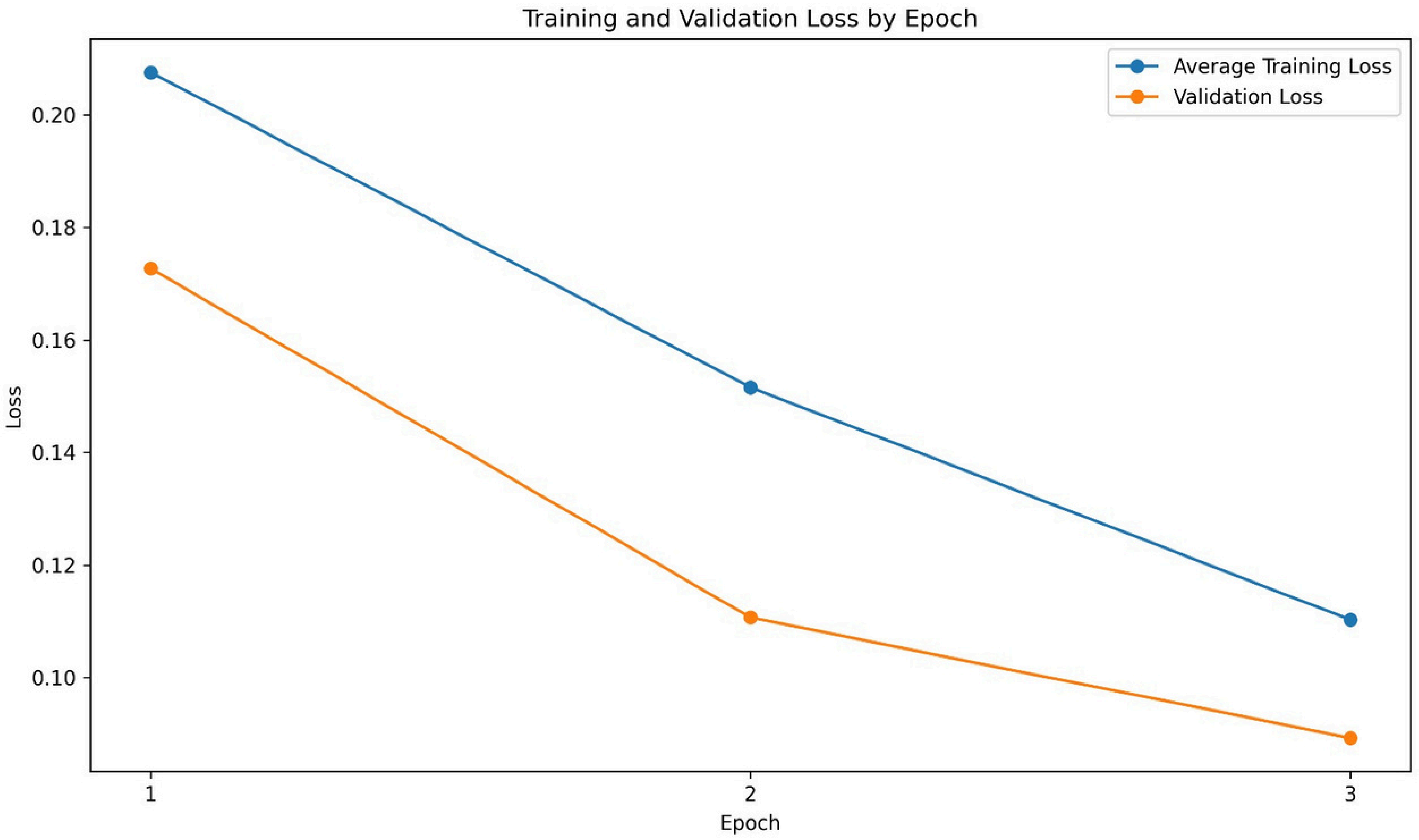
94.2%	97.6%	88.8%	97.2%	88.2%	97.9%
Ease of Use	Durability & Build Quality	Cleaning & Maintenance	Compatibility with Syringes	Storage & Handling	Performance in Fluid Delivery

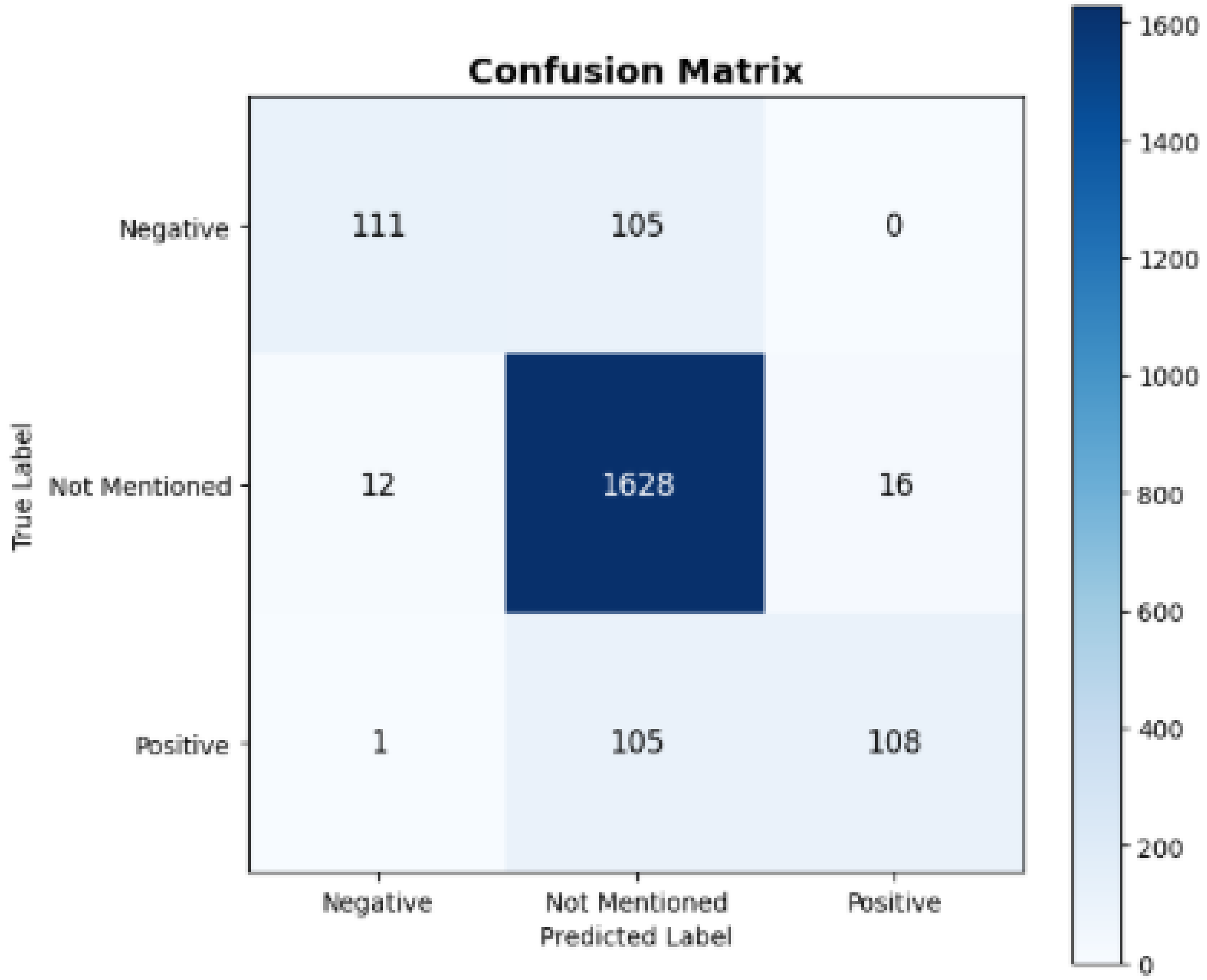


Results 2 : Medtronic 53401

Overall Performance

90.7%	46.5%	79.4%	92.6%	72.1%
Element Accuracy	Exact Vector Match	Macro F1	Precision	Recall

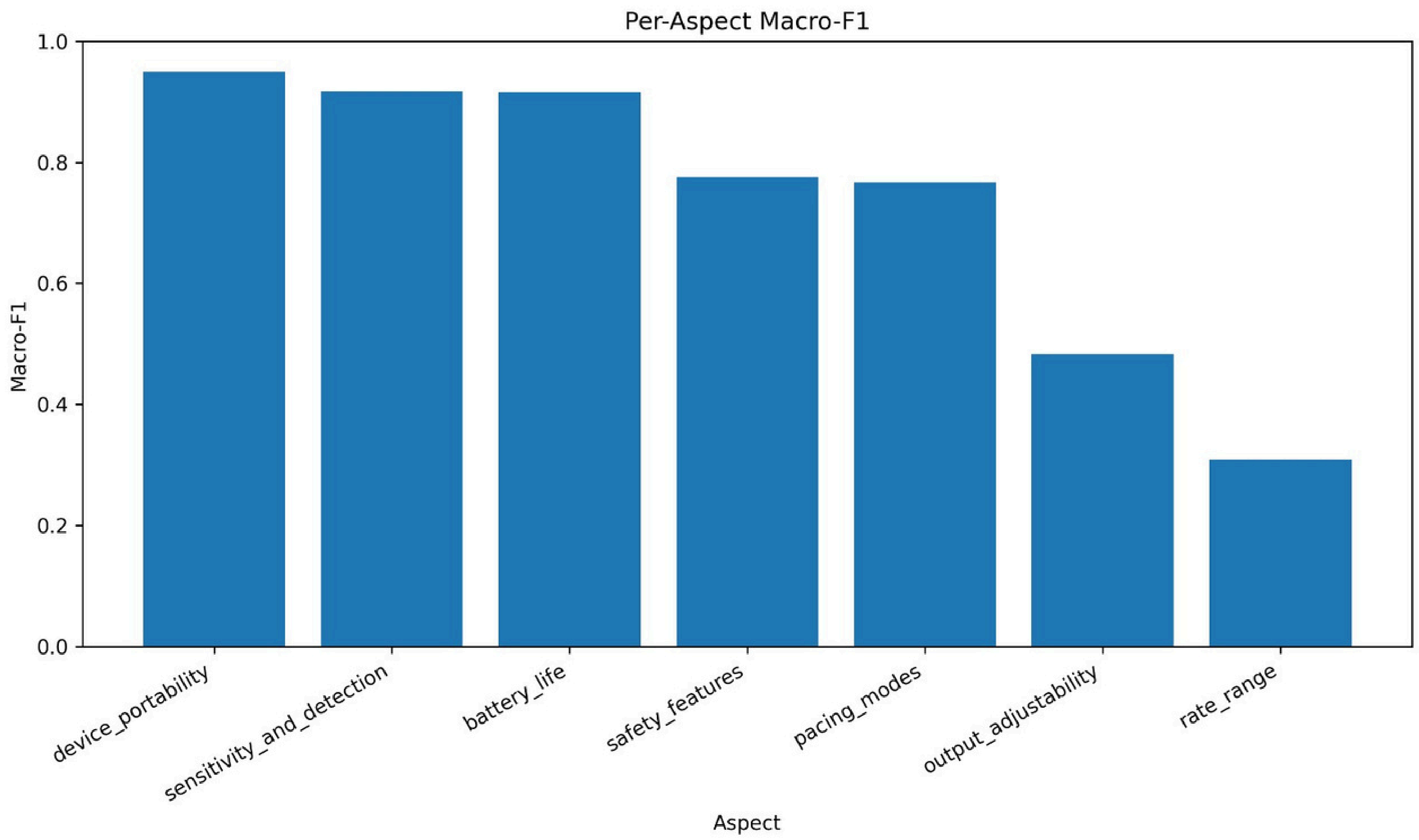




Results 2 : Medtronic 53401

Per-Aspect Macro F1

91.7%	95.0%	91.7%	76.7%	77.6%	48.3%	30.9%
Battery Life	Device Portability	Sensitivity & Detection	Pacing Modes	Safety Features	Output Adjustability	Rate Range



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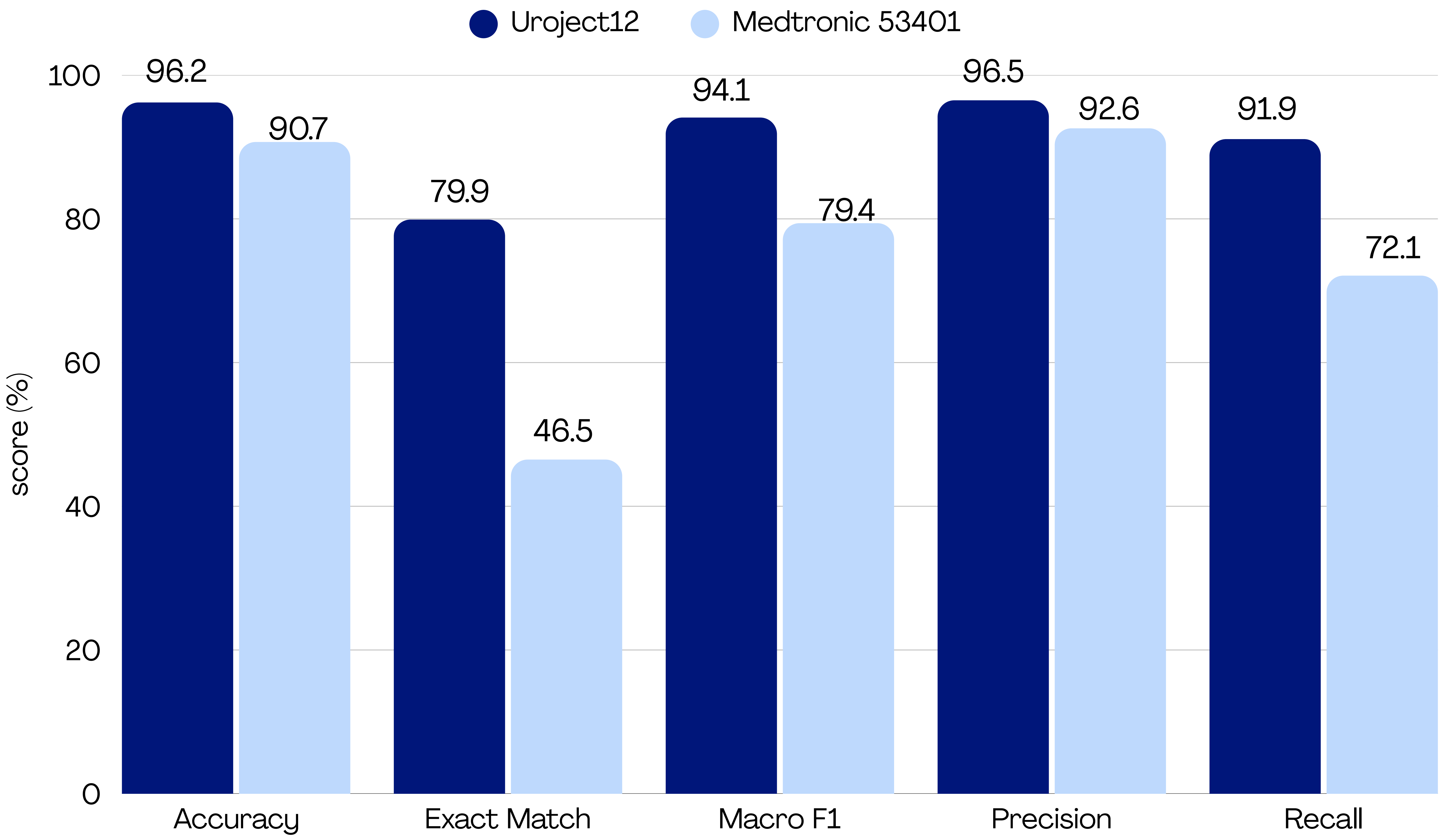

Comparison Between Models

Project Achievements & Novelty

Project Achievements

- Built an automated PDF to ABSA pipeline.
- Generated and validated high quality synthetic review datasets.
- Trained RoBERTa and DeBERTa models for aspect level sentiment prediction.
- Demonstrated the approach on two different medical devices.

Novelty

- Automatic aspect extraction from medical device PDFs.
- Synthetic review generation with sentiment and nuance control.
- Automated consistency validation.
- End to end PDF to aspect-scoring pipeline.
- Comparative evaluation of transformer architectures.

Conclusions

- RoBERTa learned the connection between review text and aspect scores.
- DeBERTa achieved stronger overall performance than RoBERTa across most evaluation metrics.
- Main errors were between mentioned and not mentioned, not positive vs. negative.
- Clear aspects were easier to learn than broad or technical aspects.
- The second device showed the pipeline is generic, but performance depends on aspect quality.
- The proposed pipeline generalized successfully across two different medical device domains.
- Real world generalization still requires testing on real reviews.

Future Work & Lessons Learned

Future Work

- Evaluate the model on real-world reviews.
- Improve aspect extraction to create clearer and less overlapping aspects.

Lessons Learned

- Synthetic data is useful only when supported by strong quality checks.
- Task formulation matters: Multi-Output Aspect Scoring fit our dataset better than classical ABSA.
- Clear, distinct aspects are easier to learn than overlapping or highly technical aspects.

Thank You